

Abhishek Kumar

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Education

National Institute Of Technology - Delhi

Aug 2023 – Dec 2027

Artificial Intelligence and Data Science

- CGPA: 7.22/10.0
- Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Data Science, Data Mining, Probability and Statistics, Data Structures and Algorithms, Operating Systems, Python, C/C++ etc.

Technical Skills

Languages: C, C++, Python, JavaScript

Web Development: HTML5, CSS3, JavaScript, Bootstrap, Node.js, Express.js, MongoDB, MySQL, REST APIs

Machine Learning & Data Science: NumPy, Pandas, Scikit-learn, TensorFlow, Keras, Matplotlib, Seaborn, NLTK

Tools & Platforms: Jupyter Notebook, Google Colab, Hadoop

Cloud Computing: AWS

CS Fundamentals: Data Structures, Algorithms, Object-Oriented Programming (OOP)

Experience

Optum (UnitedHealth Group)

Jun 2026 – Jul 2026

Technology Development Program Intern

- Contributed to the development of **Next-Gen PBM Automation**, an AI-driven solution to streamline prior authorization workflows.
- Implemented a **multi-agent LLM architecture** consisting of Past, Clinical, Policy, and Financial agents for automated decision support.
- Developed workflow orchestration logic to generate recommendations, automate approvals, and escalate complex cases for manual review.
- Created wireframes and process flows to visualize agent interactions and end-to-end automation pipelines.

Projects

AgriNova – Agriculture + Innovation

[github](#) | [live demo](#)

- Developed a multi-functional web application using **Streamlit** to assist in agricultural decision-making through deep learning, machine learning, and real-time data.
- Integrated a **CNN-based image classifier** (TensorFlow/Keras) for plant disease detection via leaf images.
- Designed a crop recommendation system using classification logic based on soil nutrients, weather conditions, and pH values.
- Implemented fertilizer suggestions using rule-based logic and dynamic **weather-based advisory** via the OpenWeatherMap API.
- **Tech Stack:** Python, TensorFlow-Keras, Streamlit, Pandas, OpenWeatherMap API

Roadmap Generator – Interactive Learning Path Visualizer

[github](#) | [live demo](#)

- Developed a **Streamlit** web app that generates structured roadmaps for any topic using **Google Gemini LLM**.
- Designed interactive graph-based visualization with **PyVis**, supporting customization of nodes, edges, and layouts.
- Enabled users to download generated roadmaps as **.json** or **.html** for offline exploration.
- **Stack:** Python, Streamlit, Google Generative AI (Gemini), PyVis, dotenv

Gesture-Based Image Cropping & OCR Text Transfer System

[github](#) 

- Developed a real-time **gesture-controlled system** for touchless image cropping and OCR-based text extraction.
- Implemented **hand gesture recognition** using **MediaPipe** and **OpenCV** with 21-hand landmark tracking.
- Built an **OCR pipeline** using **Tesseract OCR** with preprocessing techniques for accurate text extraction.
- Integrated **TCP socket communication** for reliable cross-device text transfer with low latency.
- **Tech Stack:** Python, OpenCV, MediaPipe, Tesseract OCR, Socket Programming